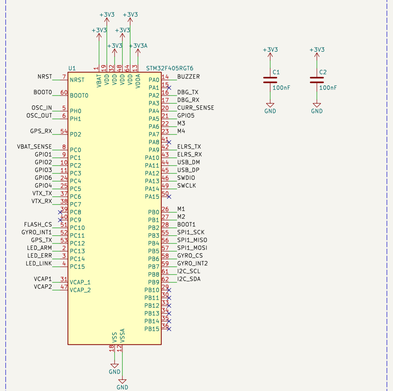
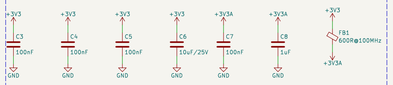
Flight Controller 5 Zoll – STM32F405 / ICM-42688-P / W25Q128

Prozessor STM32F405RGT6



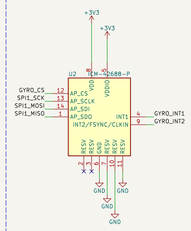


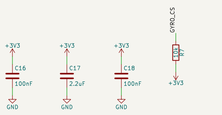




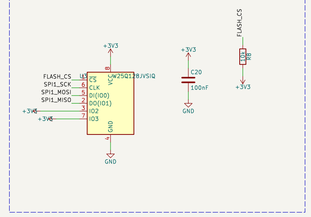


Gyro ICM-42688-P (SPI1)

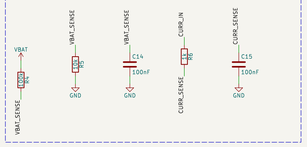




Blackbox-Flash W25Q128 (SPI1)



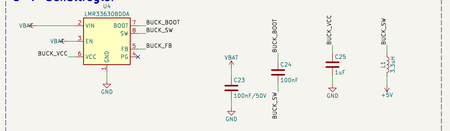
Messung Akkuspannung / Strom

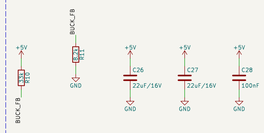


Eingang 2-6S, Verpolschutz

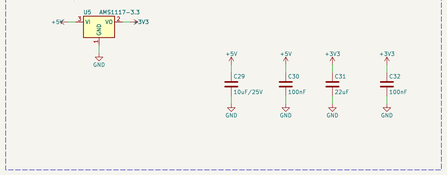


5-V-Schaltregler

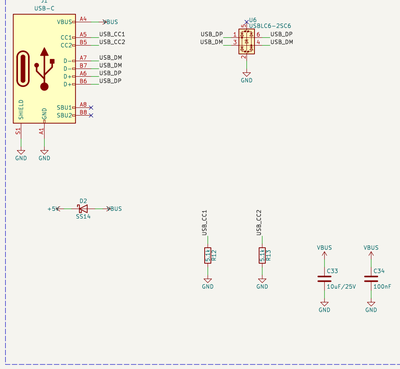




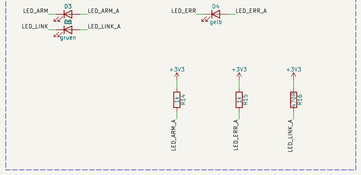
3,3-V-Regler



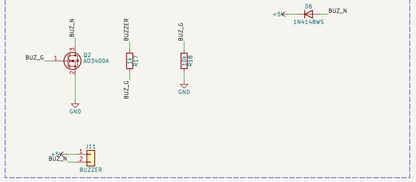
USB-C



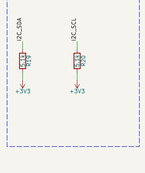
Status-LEDs



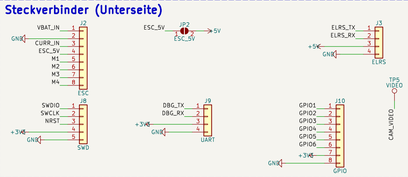
Piepser



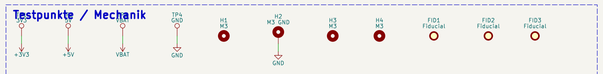
I2C Pull-ups



Steckverbinder (Unterseite)



Testpunkte / Mechanik



Versorgungsmarkierungen (PWR_FLAG) fuer die elektrische Regelpruefung



Hinweise:

- Pinbelegung MCU siehe README.md (verbindlich: TIM3 Motoren PB0/PB1/PA6/PA7, PC0 Akku, PA13/PA14 SWD, PA11/PA12 USB).
- Q1: SI2319CDS (40 V / +-20 V) statt SI2301 (nur 20 V / +-8 V, fuer 6S ungeeignet). Gleicher SOT-23-Footprint.
- U4: LMR33630B (36 V Eingang) statt MP1584/TPS54331 (28 V, zu knapp fuer 6S).
- J2 ESC: Pinfolge 1 VBAT, 2 GND, 3 CURR, 4 5V, 5–8 M1–M4 – vor dem Bestellen mit dem ESC-Kabel vergleichen!
- JP2 offen: ESC-5V-Pin ist nicht mit dem FC-5V verbunden (verhindert zwei Regler gegeneinander).
- U2 Pin 2/3 RESV laut Referenzdesign frei, 7/10/11 an GND, C19 an Pin 2 nur wegen der Spezifikation.

42 x 41 mm, 4 Lagen, Lochbild 30,5 mm Erzeugt aus hardware/fc/scripts/design.py	
Sheet: /	
File: fc.kicad_sch	
Title: Flight Controller 5 Zoll	
Size: A1	Date: 2026-10-13
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: 1.0** Id: 1/1